Probability Theory and Statistics
Lecture 12

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So far: we dealt with the distribution of a single continuous random variable: distribution function, density, expectation, distribution transforms, etc.

Now: the joint distribution of several continuous random variables (analogous to the discrete joint case) $\rightarrow$ random vectors.

Do not be scared by the n-dimensional definitions/statements below. Sometimes we will need the general n case, but in practice most often several=2, and the two-dimensional case is simpler.

Definition

Let $X_1, \ldots, X_n$ be random variables defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for some positive integer n. Then the map

$$\underline{X} = (X_1, \ldots, X_n) \colon \Omega \rightarrow \mathbb{R}^n$$

is called a random vector.

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Definition

The (joint) distribution function of $\underline{X}$ is the

$$F_{\underline{X}} : \mathbb{R}^n \to [0,1], \qquad F_{\underline{X}}(x_1, \ldots, x_n) = \mathbb{P}(X_1 < x_1, \ldots, X_n < x_n),$$

a scalar-valued function of a vector argument.

Properties of the Joint Distribution Function

Reminder: for one random variable X, the distribution function F_X satisfies:

- $\lim_{x \to -\infty} F_X(x) = 0$,
- $\lim_{x \to \infty} F_X(x) = 1$,
- F_X is left-continuous, i.e., for every $x_1 \in \mathbb{R}$, $\lim_{x \uparrow x_1} F_X(x) = F_X(x_1)$,
- F_X is nondecreasing: if $x, y \in \mathbb{R}$ and $x \leq y$, then $F_X(x) \leq F_X(y)$.

What are the analogues of these properties for a random vector?

Properties of the Joint Distribution Function

If $F_{\underline{X}}$ is the joint distribution function of a random vector $\underline{X}$, then

- $\lim_{x_1 \to -\infty, x_2 \to -\infty, \ldots, x_n \to -\infty} F_{\underline{X}}(x_1, \ldots, x_n) = 0$,
- $\lim_{x_1 \to \infty, x_2 \to \infty, \ldots, x_n \to \infty} F_{\underline{X}}(x_1, \ldots, x_n) = 1$,
- $F_{\underline{X}}$ is "(partially) left-continuous in each variable", i.e., for any $i \in \{1, \ldots, n\}$, any $x_i \in \mathbb{R}$, and fixed $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n \in \mathbb{R}$,

$$\lim_{x \uparrow x_i} F_{\underline{X}}(x_1, \ldots, x_{i-1}, x, x_{i+1}, \ldots, x_n) = F_{\underline{X}}(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n).$$

- If all but one variables are fixed, then it is (partially) nondecreasing. Namely, for $i \in \{1, \ldots, n\}$ and given $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n \in \mathbb{R}$, for any $x \leq y$,

$$F_{\underline{X}}(x_1, \ldots, x_{i-1}, x, x_{i+1}, \ldots, x_n) \leq F_{\underline{X}}(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_n).$$

Properties of the Joint Distribution Function

Discrete case $\rightarrow$ joint probability table, marginals.

$Y \backslash X$	0	1	p_Y
0	$\frac{1}{4}$	0	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
2	0	$\frac{1}{4}$	$\frac{1}{4}$
p_X	$\frac{1}{2}$	$\frac{1}{2}$	1

In n dimensions it is the same, just with an n-dimensional joint mass function (harder to draw a table).
To obtain the marginal of one variable, sum over all the other variables.

Continuous Case?

Continuous case $\to$ what do we even mean by "continuous" here?

Example ($n = 2$): On the plane, choose a point uniformly at random on the segment of the line $y = x$ between $(0, 0)$ and $(1, 1)$; denote it by (X_1, X_2).

Clearly $\mathbb{P}(X_2 = X_1) = 1$. Then X_1 and X_2 are each uniform on $[0, 1]$, so individually X_1 and X_2 are continuous random variables. However, the range of the random vector (X_1, X_2) is a subset of the one-dimensional subspace $\{(x, y) \in \mathbb{R}^2 : y = x\} \leq \mathbb{R}^2$.

We will see: (X_1, X_2) is not a jointly continuous random vector.
A jointly continuous 2-dimensional random vector (X, Y) always has a range with positive area.
It has a joint density $f_{(X,Y)} : \mathbb{R}^2 \to [0, \infty)$ whose double integral over all of $\mathbb{R}^2$ is 1.

Joint Density

Definition

Let $\underline{X} = (X_1, \ldots, X_n)$ *be a random vector. A function* $f_{\underline{X}} : \mathbb{R}^n \to [0, \infty)$ *is called the (joint) density of* $\underline{X}$ *if* $f_{\underline{X}}$ *has an improper Riemann integral over* $\mathbb{R}^n$*, and*

$$\int_{-\infty}^{x_n} \cdots \int_{-\infty}^{x_1} f_X(z_1, \ldots, z_n)\,\mathrm{d}z_1 \cdots \mathrm{d}z_n = F_X(x_1, \ldots, x_n) = \mathbb{P}(X_1 < x_1, \ldots, X_n < x_n).$$

We call X *continuous if it has a joint density.*

Remarks:

- In one dimension this is the same notion as continuity of the distribution.
- For a 2D vector $X = (X_1, X_2)$ we often write f_{X_1, X_2} instead of $f_X = f_{(X_1, X_2)}$.

Joint Density

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$$\begin{aligned}\int_{-\infty}^{x_n} \cdots \int_{-\infty}^{x_1} f_X(z_1, \dots, z_n)\,\mathrm{d}z_1 \cdots \mathrm{d}z_n &= F_X(x_1, \dots, x_n) \\ &= \mathbb{P}(X_1 < x_1, \dots, X_n < x_n).\end{aligned}$$

We call X continuous if it has a joint density.

Simplest example: choose a point $Y = (Y_1, Y_2)$ uniformly at random in the open unit square $(0,1)^2 \subset \mathbb{R}^2$. Its density is

$$f_Y(y_1, y_2) = \begin{cases} 1, & \text{if } 0 < y_1 < 1 \text{ and } 0 < y_2 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Joint Density

Theorem

Let $\underline{X} = (X_1, \ldots, X_n)$ *be a random vector. If* X *is continuous, then the function*

$$f_X(x_1, \ldots, x_n) = \begin{cases} \partial_{x_1} \cdots \partial_{x_n} F_X(x_1, \ldots, x_n), & \text{if this exists,} \\ 0, & \text{otherwise,} \end{cases}$$

is a density of X. *(The partial derivatives may be taken in any order.)*

Remark: Unlike in the one-variable statement, here the existence of a density (i.e., joint continuity of X) is an assumption, not a consequence.

Recall the earlier example with $X_2 = X_1$: $F_{(X_1, X_2)}$ is twice (totally) differentiable except along a few lines, yet $\partial_{x_1}\partial_{x_2} F_{(X_1,X_2)}(x_1, x_2)$ is 0 wherever it exists, which cannot be a joint density. (Clear?)
Thus (X_1, X_2) is not continuous, even though the marginals of X_1 and X_2 are continuous.

Joint Density

Characterization of joint densities:

Theorem

Let $f : \mathbb{R}^n \to [0, \infty)$*. Then*

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(z_1, \ldots, z_n)\, \mathrm{d}z_1 \cdots \mathrm{d}z_n = 1$$

if and only if there exists a random vector X *whose density is* f*.*

Joint Density

In one dimension we used the density to compute, e.g., probabilities of the form $\mathbb{P}(a < X < b)$. In higher dimensions:

Theorem

Let $H \subseteq \mathbb{R}^n$ be Jordan-measurable and let $X = (X_1, \dots, X_n)$ be a continuous random vector. Then

$$\mathbb{P}(X \in H) = \int_H f_X(x) \, \mathrm{d}x.$$

E.g., in $d = 2$ dimensions, for a random vector $X = (X_1, X_2)$ and a rectangle $H = (a, b) \times (c, d)$,

$$\begin{aligned} \mathbb{P}\big((X_1, X_2) \in H\big) &= \mathbb{P}\big(a < X_1 < b,\ c < X_2 < d\big) \\ &= \int_a^b \int_c^d f_{(X_1,X_2)}(x_1, x_2) \, \mathrm{d}x_2 \, \mathrm{d}x_1 \bigg(= \int_H f_X(x) \, \mathrm{d}x \bigg). \end{aligned}$$

Example

Let $f\colon \mathbb{R}^2 \to [0,\infty)$ be

$$f(x,y) = \begin{cases} 4xy, & \text{if } 0 < x, y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- Does there exist a random vector (X, Y) with density f?
- If yes: compute $\mathbb{P}(X < Y)$.

(Solution: yes; $1/2$; see the lecture.)

Example

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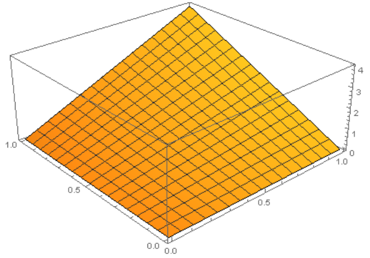
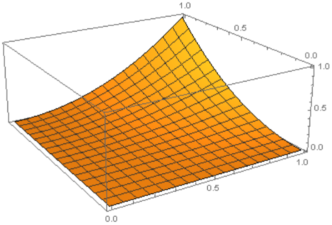
Joint density and distribution function in the example:





Marginal Distributions ($n = 2$)

On the next few slides we first consider the (less intimidating) $n = 2$ case. Afterwards we comment on general n.

If (X, Y) is a random vector, then F_X and F_Y are called the marginal distribution functions. How do we obtain them from $F_{(X,Y)}$?
In the joint distribution function, send the other coordinate to $+\infty$:

- $F_X(x) = \lim_{y\to\infty} F_{(X,Y)}(x, y)$, $x \in \mathbb{R}$,
- $F_Y(y) = \lim_{x\to\infty} F_{(X,Y)}(x, y)$, $y \in \mathbb{R}$.

(Sketch of proof: see the lecture $\to$ continuity of probability measures!)

Marginal Distributions ($n = 2$)

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- $F_Y(y) = \lim_{x\to\infty} F_{(X,Y)}(x, y)$, $y \in \mathbb{R}$.

Question: how do we obtain the marginal density of X?

- 1. approach: compute F_X and then differentiate.
- 2. approach: integrate out the joint density with respect to y:

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x, y)\,\mathrm{d}y, \qquad x \in \mathbb{R}.$$

Similarly, integrating with respect to x gives the marginal density of Y:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x, y)\,\mathrm{d}x, \qquad y \in \mathbb{R}.$$

Marginal Density ($n = 2$)

Example: choose a point $X = (X, Y)$ uniformly at random on the square $(0,1)^2$:

$$f_X(x, y) = \begin{cases} 1, & \text{if } 0 < x < 1 \text{ and } 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Thus

$$f_X(x) = \int_{-\infty}^{\infty} f_X(x, y) \, \mathrm{d}y = \int_0^1 1 \, \mathrm{d}y = 1$$

for $x \in (0,1)$ and 0 otherwise.
Similarly $f_Y(y) = 1$ for $y \in (0,1)$ and 0 otherwise.
Both X and Y have uniform marginals on $(0,1)$.

Marginal Density ($n = 2$)

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,\mathrm{d}y, \qquad x \in \mathbb{R}.$$

Why is this true? $\rightarrow$ We saw:

$$F_X(x) = \mathbb{P}(X < x) = \mathbb{P}(-\infty < X < x,\ -\infty < Y < \infty) =$$

$$= \int_{-\infty}^{x} \int_{-\infty}^{\infty} f_{(X,Y)}(t,y)\,\mathrm{d}y\,\mathrm{d}t.$$

Hence,

$$f_X(x) = F_X'(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y.$$

This explanation is valid if $F_{(X,Y)}$ is twice totally differentiable everywhere (hence all second-order mixed partials exist).
In the general case we obtain the same formula using substitution in integrals instead of differentiating.

Exercise

$$f_{X,Y}(x,y) := \begin{cases} axy^2, \text{ if } 1 < x < 2 \text{ and } 1 < y < 3 \\ 0 \text{ otherwise} \end{cases}$$

1. $a = ?$
2. $F_{X,Y} = ?$
3. Calculate F_X, F_Y (marginal distributions)
4. Calculate f_X, f_Y (marginal densities)
5. Calculate $\mathbb{E}(X)$ and $\mathbb{E}(Y)$!